Roll No. | | | | | | | | | | |

Total No. of Questions : 13

Total No. of Pages : 02

B.Pharmacy (Sem.-6)

MEDICINAL CHEMISTRY-III-THEORY

Subject Code : BP601T

M.Code : 77986

Date of Examination : 30-04-2025

Time : 3 Hrs.

Max. Marks : 75

INSTRUCTIONS TO CANDIDATES :

1. SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.
2. SECTION-B contains THREE questions carrying TEN marks each and students have to attempt any TWO questions.
3. SECTION-C contains NINE questions carrying FIVE marks each and students have to attempt any SEVEN questions.

SECTION - A

1. **Briefly explain :**

   a) Give structure and IUPAC name of any one Folate reductase inhibitor.

   b) Write down various steps involved in the synthesis of chloramphenicol.

   c) Write down the moa of acyclovir. Give its structure.

   d) Name any four antifungal antibiotics.

   e) Give the synthesis of Dapsone.

   f) Differentiate between structure based drug design and ligand based drug design approach.

   g) What are prodrugs? Give example.

   h) Write down the mechanism of action of ketokonazole and ofloxacin.

   i) Name any two molecular docking software's available now days.

   j) Comment upon the therapeutic uses of nitrofurantoin.

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SECTION - B

2. Classify Sulphonamides. Give examples. Comment upon their SAR and write synthesis of any one drug belonging to this class.

3. Give detailed note on different physicochemical and steric parameters used in QSAR studies.

4. Write the synthetic procedures for the following drugs :

   a) Metronidazole

   b) Pamaquine

   c) Ciprofloxacin

   d) Paraamino salycilic acid

SECTION - C

5. Write a note on drugs used as First line treatment of Tuberculosis.

6. Discuss SAR of quinolines as anti-malarial agents.

7. Classify Antiviral agents. Give examples of drugs belonging to each category.

8. Give structure, IUPAC name, synthesis and uses of Miconazole.

9. Discuss the SAR of tetracycline antibiotics.

10. Write a note on tetracyclins as antibiotic agents.

11. Write a detailed note on combinatorial chemistry.

12. Give structure, IUPAC name, moa and use of Tolnaftate.

13. Discuss various applications of Prodrug design.

NOTE : Disclosure of identity by writing mobile number or making passing request on any page of Answer sheet will lead to UMC against the Student.